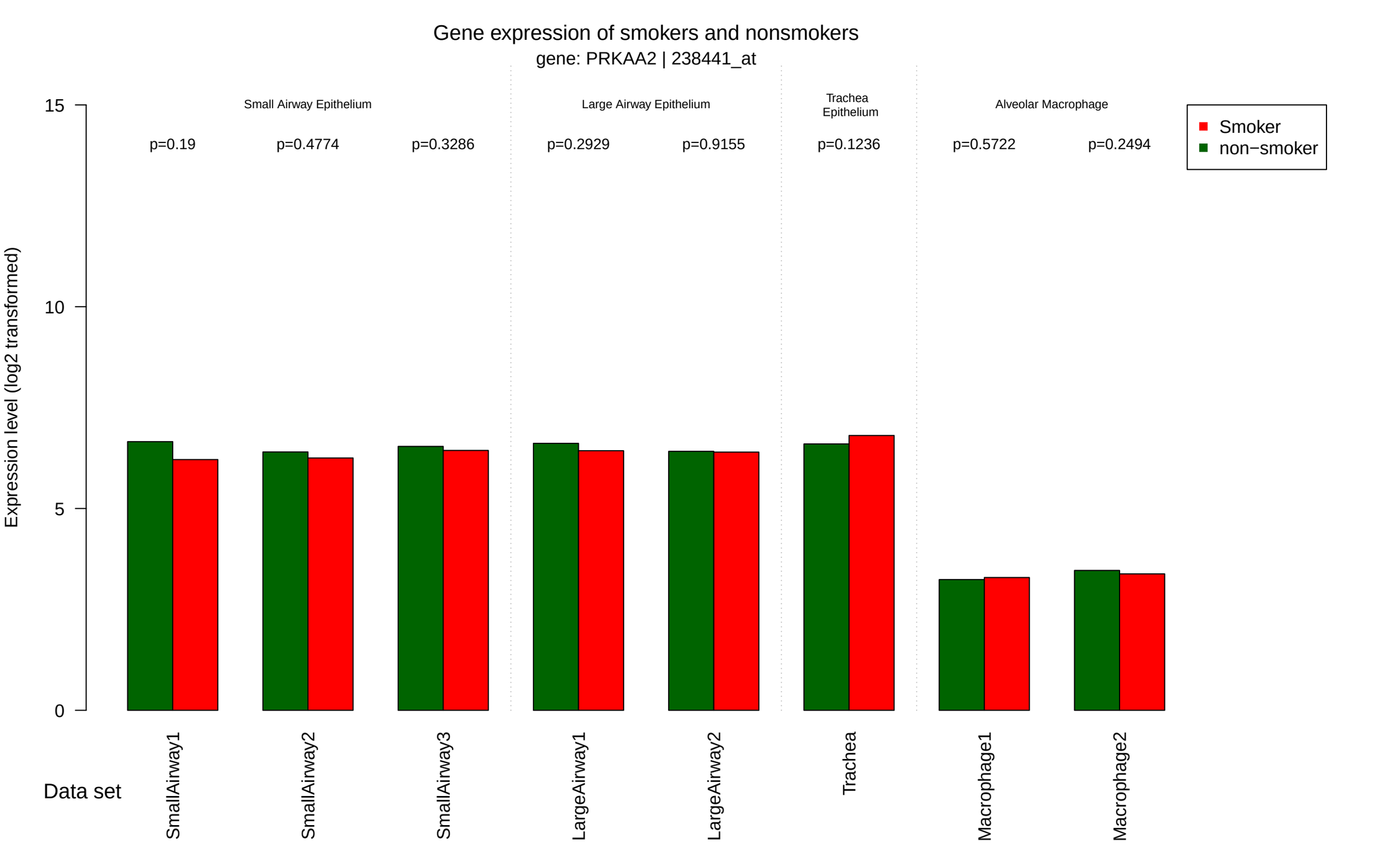
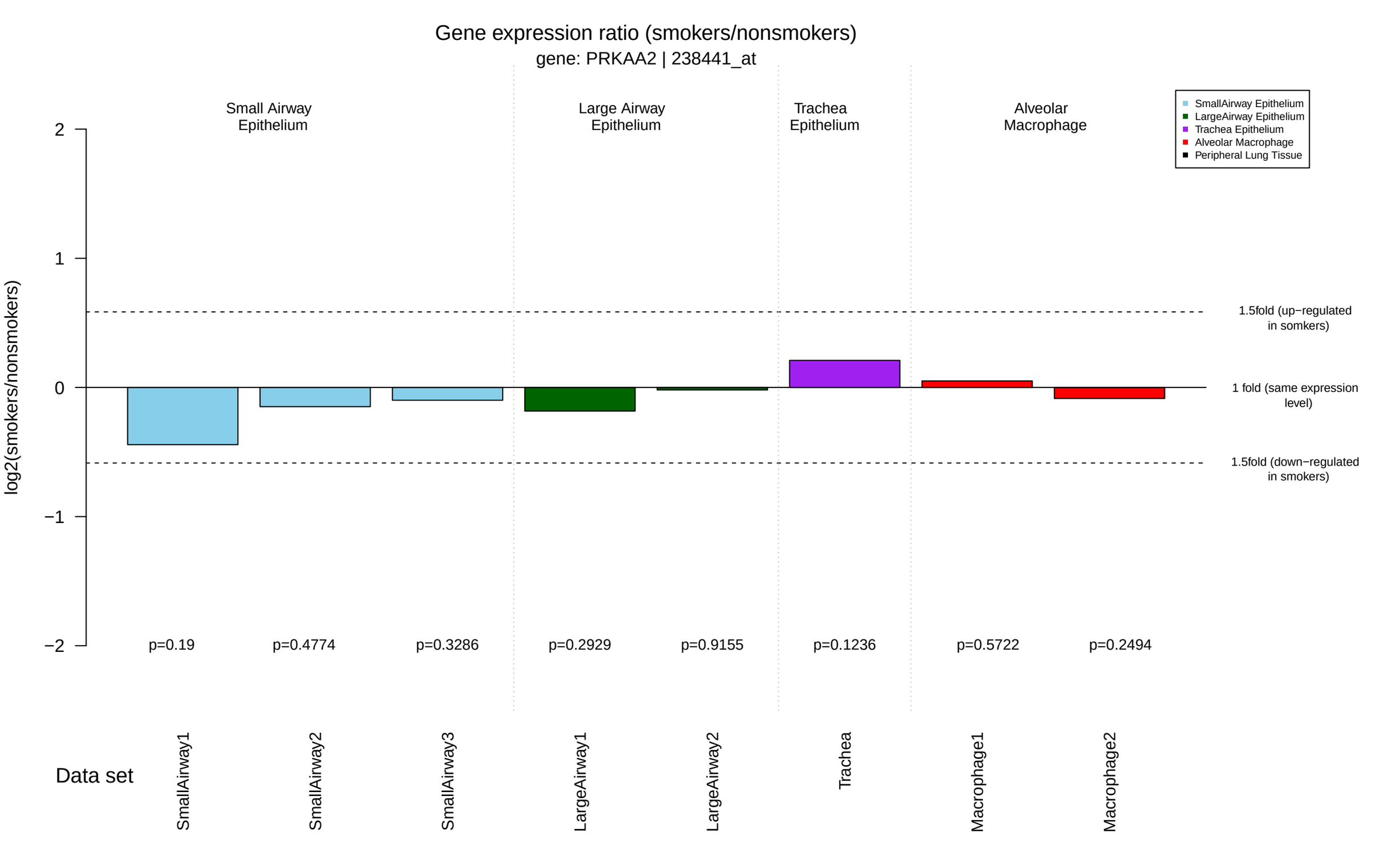
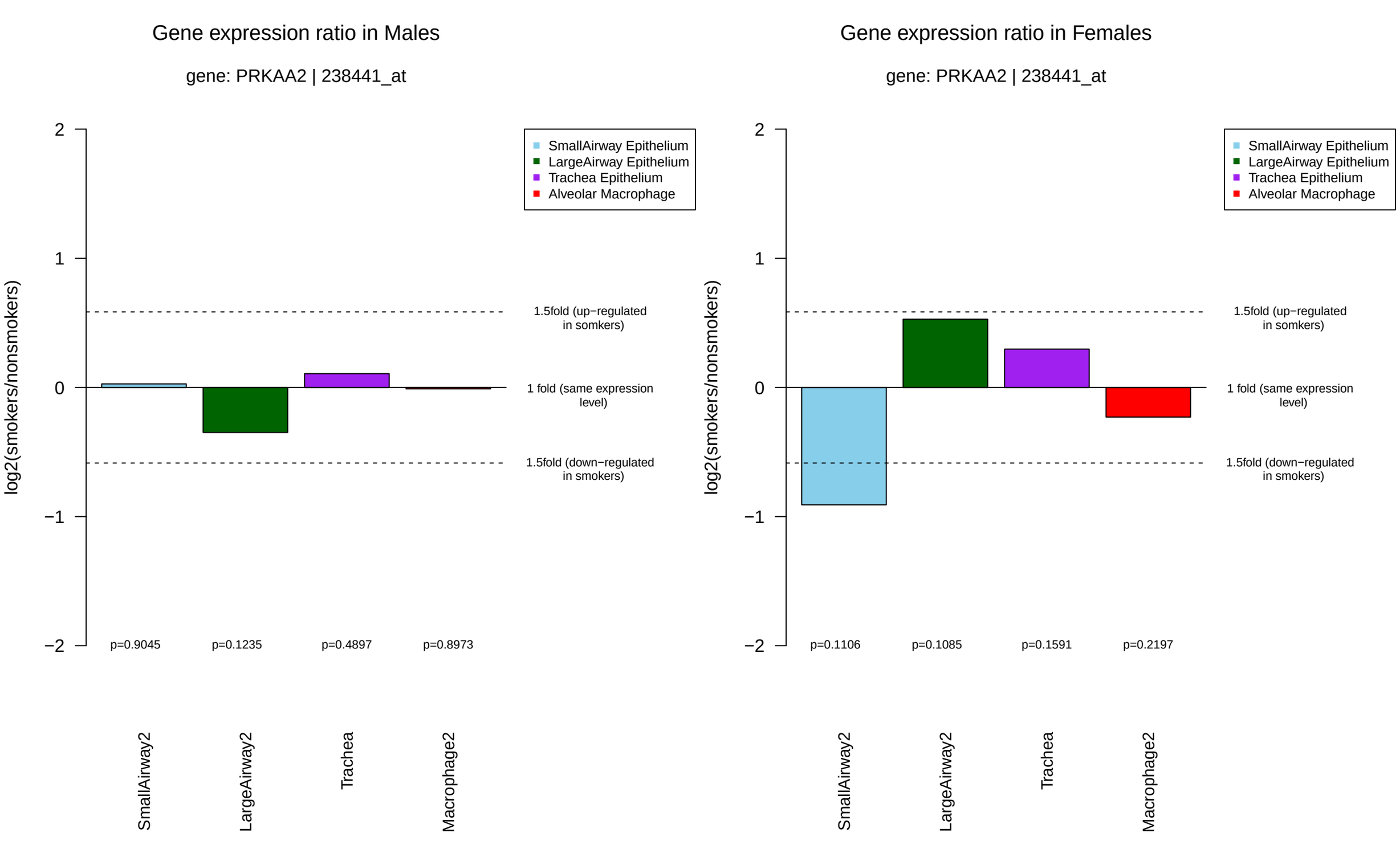
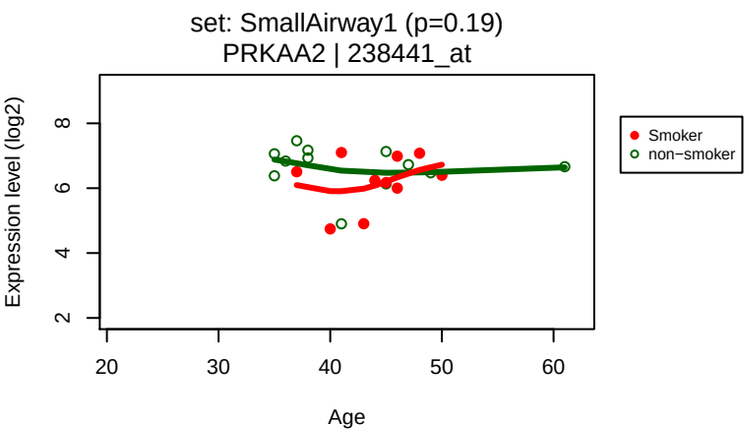
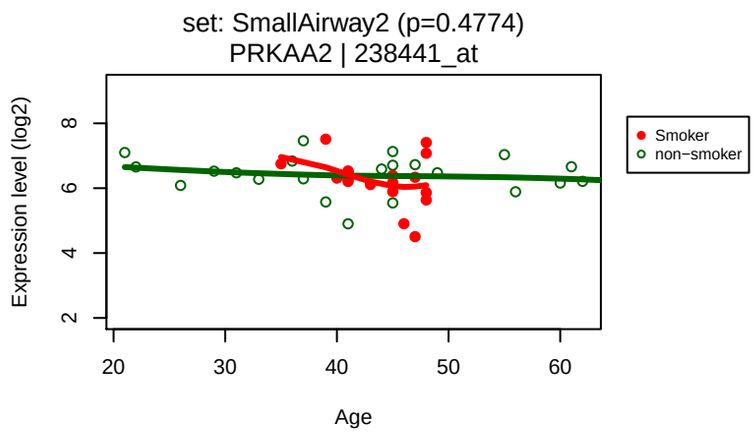
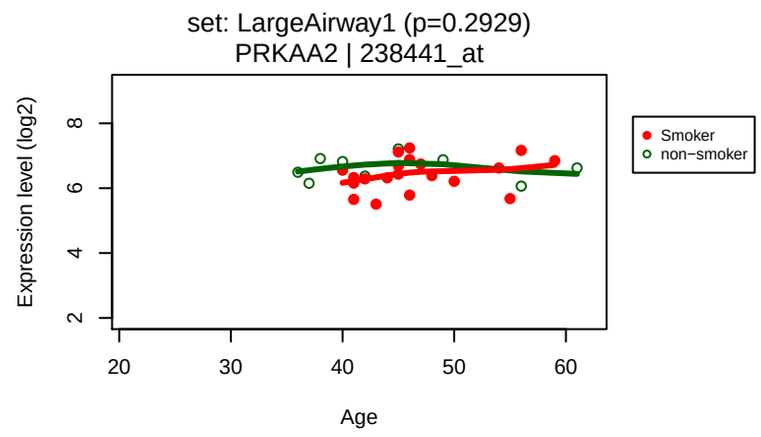

Gene expression of smokers and nonsmokers
gene: PRKAA2 | 238441_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.19
p=0.4774
p=0.3286
p=0.2929
p=0.9155
p=0.1236
p=0.5722
p=0.2494
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

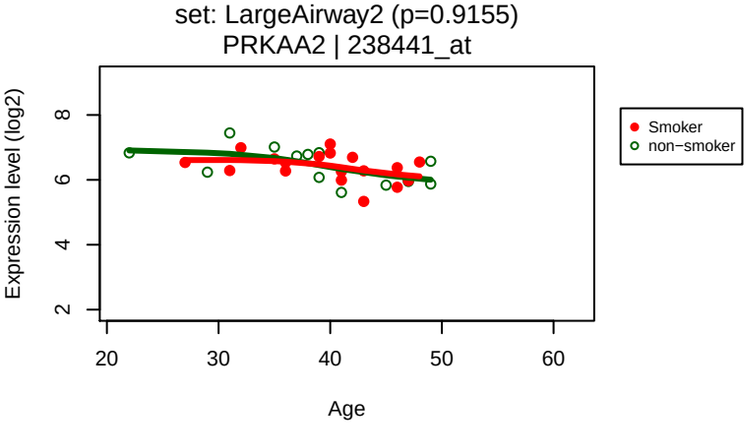

Gene expression ratio (smokers/nonsmokers)
gene: PRKAA2 | 238441_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.19
p=0.4774
p=0.3286
p=0.2929
p=0.9155
p=0.1236
p=0.5722
p=0.2494
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

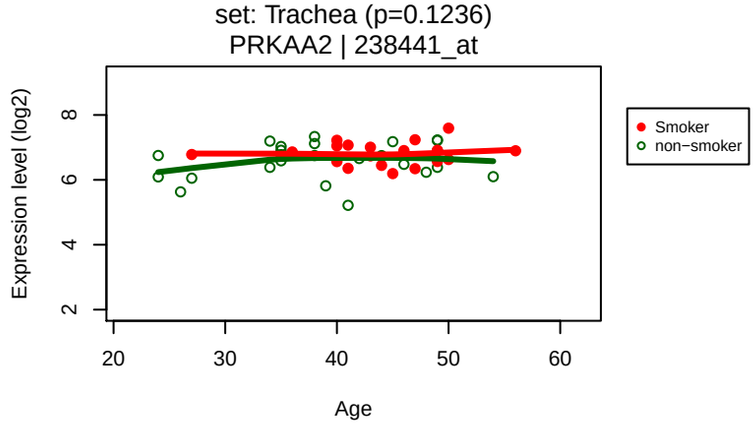

Gene expression ratio in Males
gene: PRKAA2 | 238441_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9045
p=0.1235
p=0.4897
p=0.8973
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAA2 | 238441_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1106
p=0.1085
p=0.1591
p=0.2197
SmallAirway2
LargeAirway2
Trachea
Macrophage2

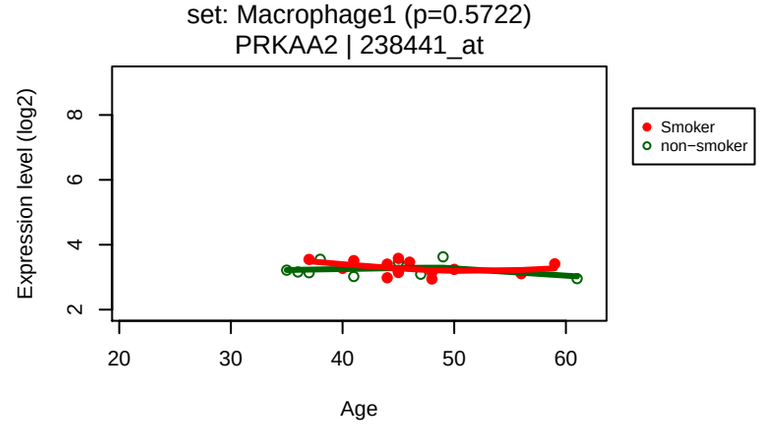

set: SmallAirway1 (p=0.19)
PRKAA2 | 238441_at
Expression level (log2)
Age
Smoker
non−smoker

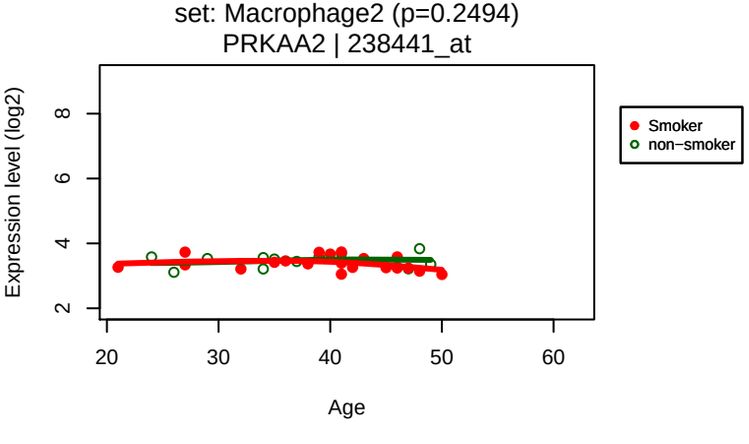

set: SmallAirway2 (p=0.4774)
PRKAA2 | 238441_at
Expression level (log2)
Age
Smoker
non−smoker

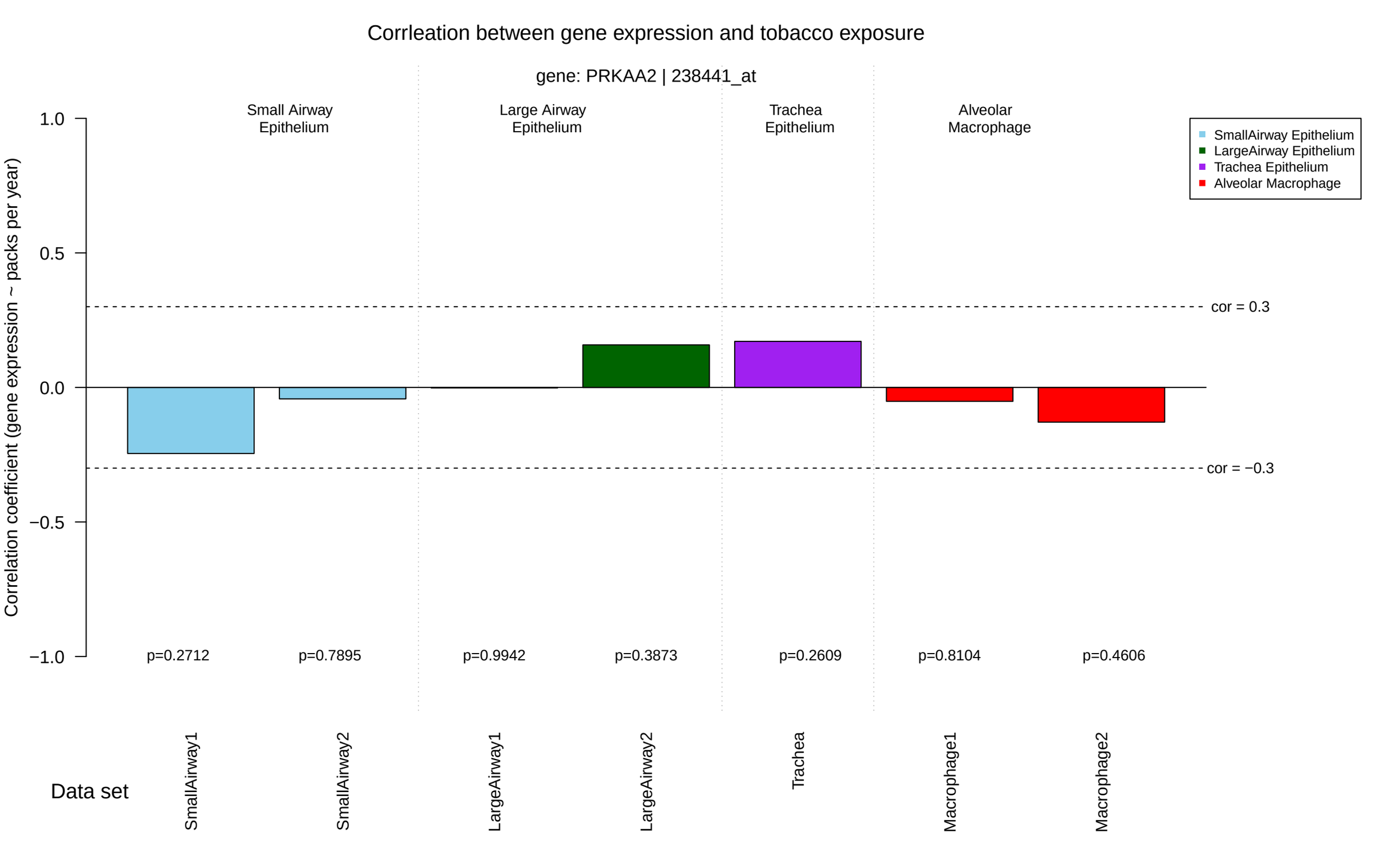

set: LargeAirway1 (p=0.2929)
PRKAA2 | 238441_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9155)
PRKAA2 | 238441_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1236)
PRKAA2 | 238441_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5722)
PRKAA2 | 238441_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2494)
PRKAA2 | 238441_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAA2 | 238441_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2712
p=0.7895
p=0.9942
p=0.3873
p=0.2609
p=0.8104
p=0.4606
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2